Class: TE **Subject: AISC** **Sem-V**

Experiment No: 09

Course Outcome: Implement Fuzzy Membership Functions
Blooms Level: L4 - Analyze

Aim: Analyze and implement different fuzzy membership functions to classify room temperature as Cold, Comfortable, or Hot. Compare the membership values obtained using triangular, trapezoidal, and Gaussian membership functions for selected temperature values.

Consider room temperature in the range **0°C to 40°C**.

Define the fuzzy sets:

- **Cold**
- **Comfortable**
- **Hot**

Implement the following membership functions:

1. Triangular
2. Trapezoidal
3. Gaussian

For selected temperatures such as:

10°C, 15°C, 20°C, 25°C, 30°C, 35°C

calculate the membership values and compare the results.

Abstract:
This experiment implements and analyzes three commonly used fuzzy membership functions: Triangular, Trapezoidal, and Gaussian using Python. The membership values are calculated for different temperature values and the functions are plotted to compare their shapes and behavior.

Sample Input and Output:

Input: 10°C, 15°C, 20°C, 25°C, 30°C, 35°C

Sample Output: A graph showing the comparison of the three membership functions

Theory:

A **fuzzy membership function** represents the degree to which an input belongs to a fuzzy set. The membership value generally ranges from **0 to 1**, where 0 means no membership and 1 means complete membership.

The three membership functions used in this experiment are:

1. **Triangular Membership Function:** A triangular membership function is defined by three parameters a, b, c:

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$$\mu(x) = \begin{cases} 0, & x \le a \\ \dfrac{x-a}{b-a}, & a < x \le b \\ \dfrac{c-x}{c-b}, & b < x < c \\ 0, & x \ge c \end{cases}$$

where:

a = left point b = peak c = right point

2. **Trapezoidal Membership Function:** Has a flat top where the membership value remains 1 and is defined using four parameters.

$$\mu(x) = \begin{cases} 0, & x \le a \\ \dfrac{x-a}{b-a}, & a < x \le b \\ 1, & b < x \le c \\ \dfrac{d-x}{d-c}, & c < x < d \\ 0, & x \ge d \end{cases}$$

where:

- a = left boundary
- b = start of the flat region
- c = end of the flat region
- d = right boundary

3. **Gaussian Membership Function:** Has a smooth bell-shaped curve and provides gradual transitions between membership values.

$$\mu(x) = e^{-\frac{(x-c)^2}{2\sigma^2}}$$

where:

- c = center of the function
- σ = standard deviation, which controls the width of the curve
- e = Euler's number

The Gaussian function produces a **smooth bell-shaped curve**, with the maximum membership value of **1 at** $x = c$.

By plotting these functions, students can **compare their shapes and analyze how differently they represent the same input range**.

Code:

https://colab.research.google.com/drive/1veBkLNiscGbnsOxQJ4xc9KLV2XKwFyV4?usp=sharing

Conclusion:

The three fuzzy membership functions were successfully implemented and plotted using Python. The comparison shows that each function provides a different way of representing the degree of membership. **Triangular and trapezoidal functions provide simple piecewise representations, while the Gaussian function provides a smoother transition.**

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Exercise 1:
A college wants to classify students based on their **attendance percentage**. The fuzzy sets are:

- **Low Attendance**
- **Moderate Attendance**
- **High Attendance**

Implement **Triangular, Trapezoidal, and Gaussian membership functions** for attendance values ranging from **0% to 100%**.

Task

1. Decide suitable parameters for the three membership functions.
2. Calculate membership values for: 40%, 50%, 60%, 70%, 80%, 90%
3. Plot the three membership functions.
4. Compare the graphs.
5. Identify which membership function is more suitable for representing **High Attendance** and explain why.

Exercise:
An air conditioner needs to classify room temperature into:

- **Cool**
- **Comfortable**
- **Warm**

The temperature range is **15°C to 40°C**.
Implement and compare **Triangular, Trapezoidal, and Gaussian membership functions**.

Task

1. Choose suitable parameters for each membership function.
2. Calculate membership values for:
   18°C, 22°C, 25°C, 28°C, 32°C, 36°C
3. Plot the three membership functions.
4. Compare the shapes and membership values.
5. Decide which membership function would be **most suitable for the Comfortable temperature category**.
6. Give a short justification for your choice,